Q $y = \tan x$ $\quad y'?$

A $y = \frac{\sin x}{\cos x}$ $\quad$ (Not y')

$$y' = \frac{(\cos x)(\sin x)' - (\sin x)(\cos x)'}{(\cos x)^2} = \frac{(\cos x)(\cos x) - (\sin x)(-\sin x)}{\cos^2 x}$$

$$= \frac{\cos^2 x + \sin^2 x}{\cos^2 x}$$

$$= \frac{1}{\cos^2 x}$$

$$= \sec^2 x$$